

GMO IM.E.K. FEDOROVA, Russian Federation

WMO: 202920

Lat: 77.713N

Lon: 104.292E

Elev: 15

StdP: 101.14

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-39.6	-37.3	-41.4	0.1	-39.4	-39.3	0.1	-37.1	18.8	-17.4	16.7	-19.6	3.4	250	1.200

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	3.0	9.0	7.8	6.5	5.6	4.6	3.9	7.8	8.8	5.7	6.5	3.9	4.5	5.3	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7.1	6.2	8.4	4.9	5.4	6.3	3.3	4.8	4.3	24.6	8.9	20.1	6.6	16.6	4.6	17.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.4	13.6	12.2	DB	-39.8	13.2	3.5	3.3	-42.4	15.6	-44.4	17.6	-46.4	19.4	-49.0	21.8	(4)
(5)			WB	-39.9	10.9	3.8	2.5	-42.6	12.7	-44.9	14.2	-47.0	15.6	-49.8	17.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-13.0	-26.0	-26.7	-25.1	-18.5	-9.5	-0.9	1.5	1.7	-1.4	-9.2	-18.8	-23.8
	DBStd	12.00	6.51	6.32	6.77	5.94	4.64	2.30	2.04	2.46	3.19	5.96	5.64	6.09
	HDD10.0	8389	1117	1026	1087	856	604	326	263	259	343	596	863	1049
	HDD18.3	11430	1376	1259	1346	1106	862	576	521	517	593	854	1113	1307
	CDD10.0	1	0	0	0	0	0	0	1	1	0	0	0	0
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	5.9	6.1	6.3	5.5	5.8	5.6	5.8	5.8	5.7	6.1	6.0	5.9	
Precipitation	PrecAvg	218	12	13	14	11	13	19	31	33	24	20	13	14
	PrecMax	362	59	50	66	35	50	50	76	97	56	57	38	53
	PrecMin	89	1	2	0	0	3	2	2	5	7	2	2	2
	PrecStd	60	10	9	12	8	9	12	20	21	13	11	8	11
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-4.7	-7.6	-4.3	-0.1	1.6	7.3	13.4	13.1	7.9	2.3	-1.4	-4.2
		MCWB	-5.2	-7.9	-4.7	-0.6	0.7	6.0	11.2	11.0	7.1	1.6	-2.0	-4.5
	2%	DB	-9.7	-11.9	-7.3	-4.1	-0.1	4.1	8.9	9.5	5.8	0.7	-4.1	-8.2
		MCWB	-10.1	-12.1	-7.6	-4.7	-0.6	2.8	7.5	8.4	5.2	0.2	-4.5	-8.6
	5%	DB	-14.0	-14.9	-11.3	-7.3	-1.6	2.7	5.9	7.0	4.0	-0.6	-8.5	-12.7
		MCWB	-14.2	-15.2	-11.6	-7.8	-2.1	1.8	4.9	6.2	3.5	-1.1	-9.0	-13.1
	10%	DB	-17.5	-17.8	-15.2	-10.1	-2.9	1.6	3.8	4.9	2.2	-1.7	-11.0	-15.7
		MCWB	-17.8	-18.1	-15.5	-10.5	-3.4	0.8	3.0	4.3	1.8	-2.2	-11.4	-16.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-4.9	-7.8	-4.6	-0.7	0.9	6.4	11.2	11.1	7.3	1.6	-1.9	-4.6
		MCDB	-4.9	-7.6	-4.4	-0.1	1.5	7.4	13.0	12.5	7.9	2.0	-1.5	-4.2
	2%	WB	-10.1	-12.1	-7.7	-4.6	-0.6	2.9	7.6	8.5	5.1	0.2	-4.5	-8.5
		MCDB	-9.8	-12.0	-7.3	-4.1	-0.2	3.7	8.8	9.3	5.6	0.6	-4.0	-8.2
	5%	WB	-14.2	-15.2	-11.5	-7.8	-2.1	1.9	4.9	6.3	3.5	-1.0	-8.9	-13.0
		MCDB	-14.0	-15.0	-11.2	-7.4	-1.7	2.6	5.9	7.0	3.9	-0.6	-8.6	-12.7
	10%	WB	-17.7	-18.1	-15.5	-10.5	-3.4	1.0	3.1	4.3	1.9	-2.1	-11.4	-15.9
		MCDB	-17.5	-17.8	-15.2	-10.1	-2.9	1.4	3.7	4.8	2.2	-1.7	-11.0	-15.6
Mean Daily Temperature Range	5% DB	MDBR	5.6	5.3	5.7	5.6	4.1	2.5	2.8	3.0	2.6	4.5	5.4	5.4
		MCDBR	8.9	8.3	7.6	6.9	3.9	4.3	7.7	7.4	4.1	3.7	6.7	8.5
		MCWBR	8.9	8.2	7.6	6.8	3.8	3.5	6.5	6.2	3.6	3.7	6.7	8.4
	5% WB	MCDBR	8.9	8.3	7.6	7.0	4.0	3.9	7.6	7.3	3.9	3.3	6.7	8.6
		MCWBR	8.9	8.2	7.6	6.8	3.9	3.4	6.5	6.3	3.6	3.4	6.7	8.5
Clear-Sky Solar Irradiance	taub	N/A	0.172	0.255	0.294	0.322	0.312	0.349	0.326	0.266	0.182	N/A	N/A	
	taud	N/A	1.873	1.975	2.042	2.058	2.261	2.305	2.397	2.293	1.937	N/A	N/A	
	Ebn at Noon	0	64	543	740	790	833	772	729	587	38	0	0	
	Edh at Noon	0	6	57	96	115	99	91	70	44	4	0	0	
All-Sky Solar Radiation	RadAvg	0.00	0.03	0.96	3.49	5.74	6.07	4.81	2.69	0.95	0.13	0.00	0.00	
	RadStd	0.00	0.00	0.10	0.19	0.28	0.56	0.30	0.29	0.10	0.02	0.00	0.00	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page